Blockchain based systems and a new order

BGIN #1 Tutorial Session
November 23, 2020

Director for International Digital Strategy and Policy,
Financial Services Agency Japan

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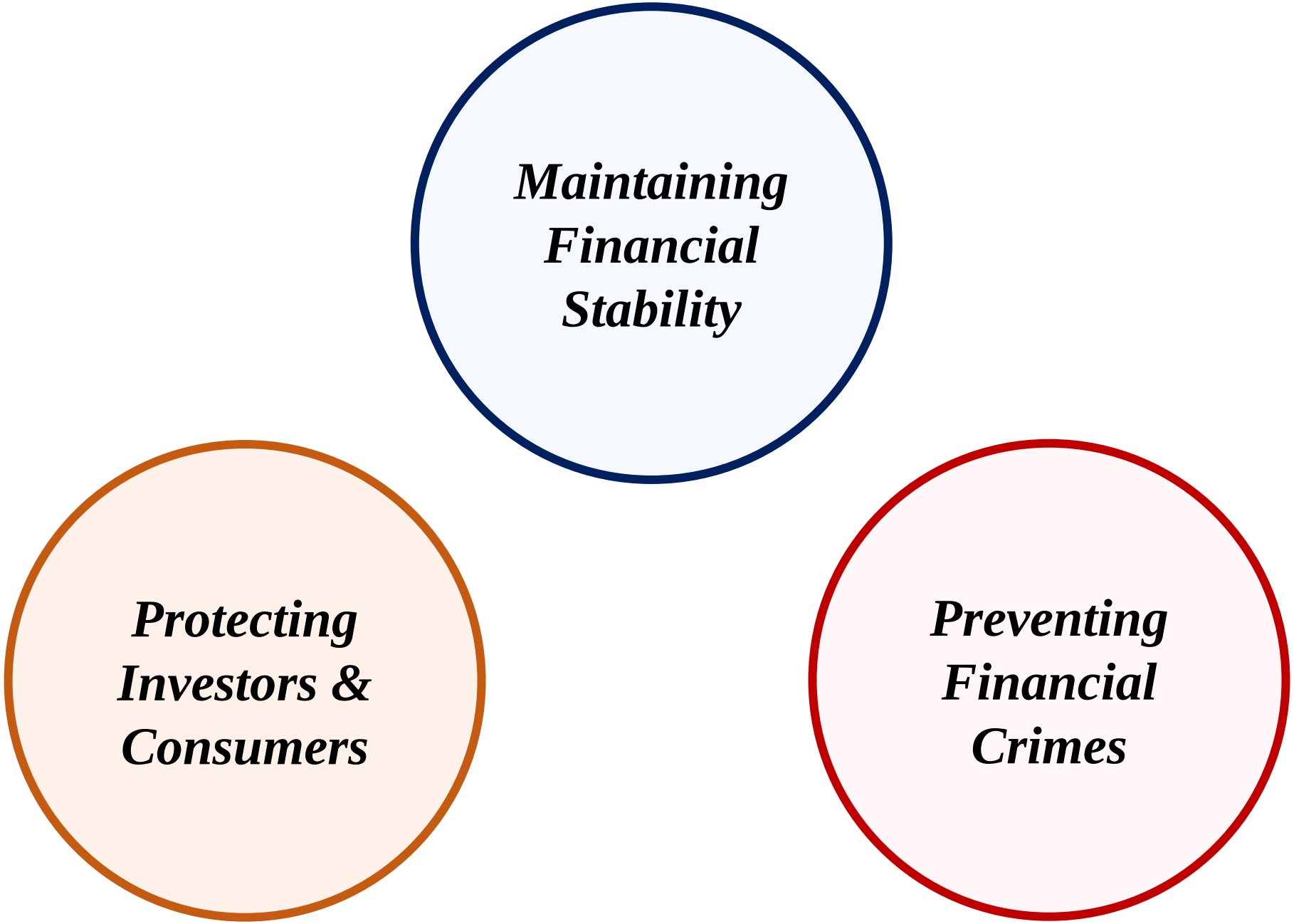
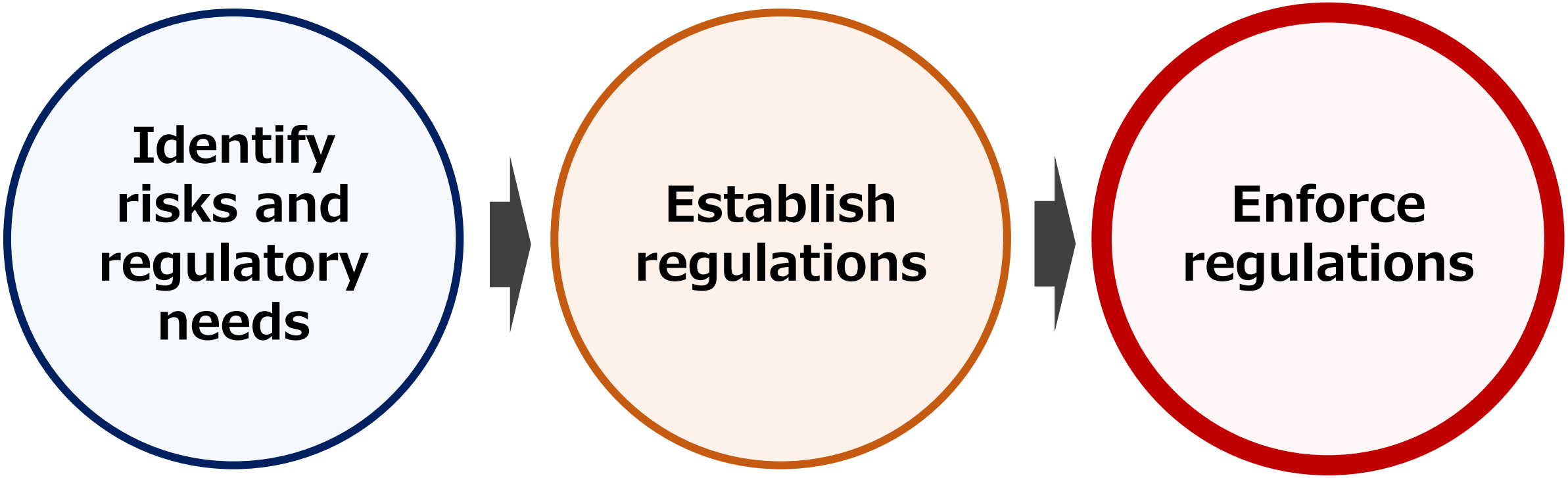
(Opinions presented here belong to myself and don' represent organizations to which I belong.)

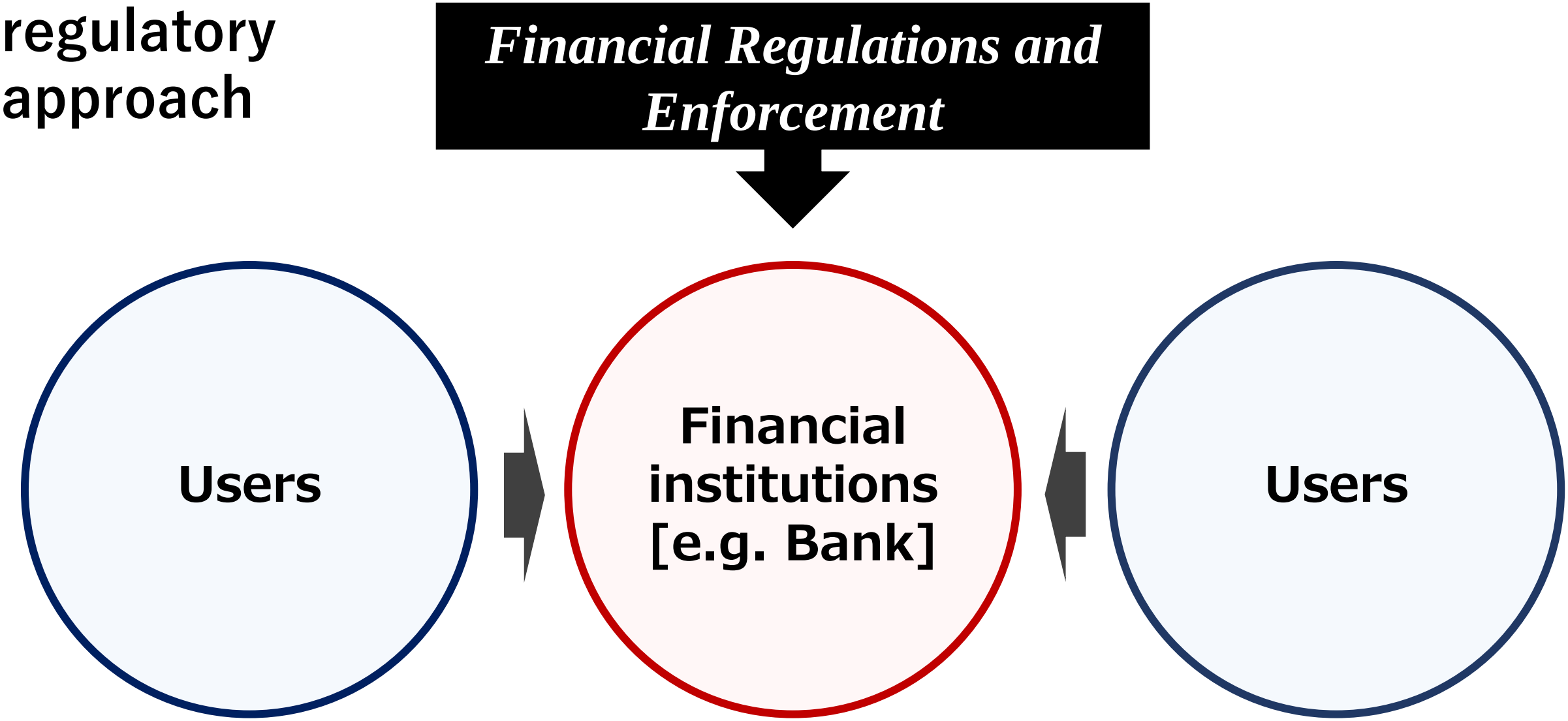
Financial Regulatory Goals



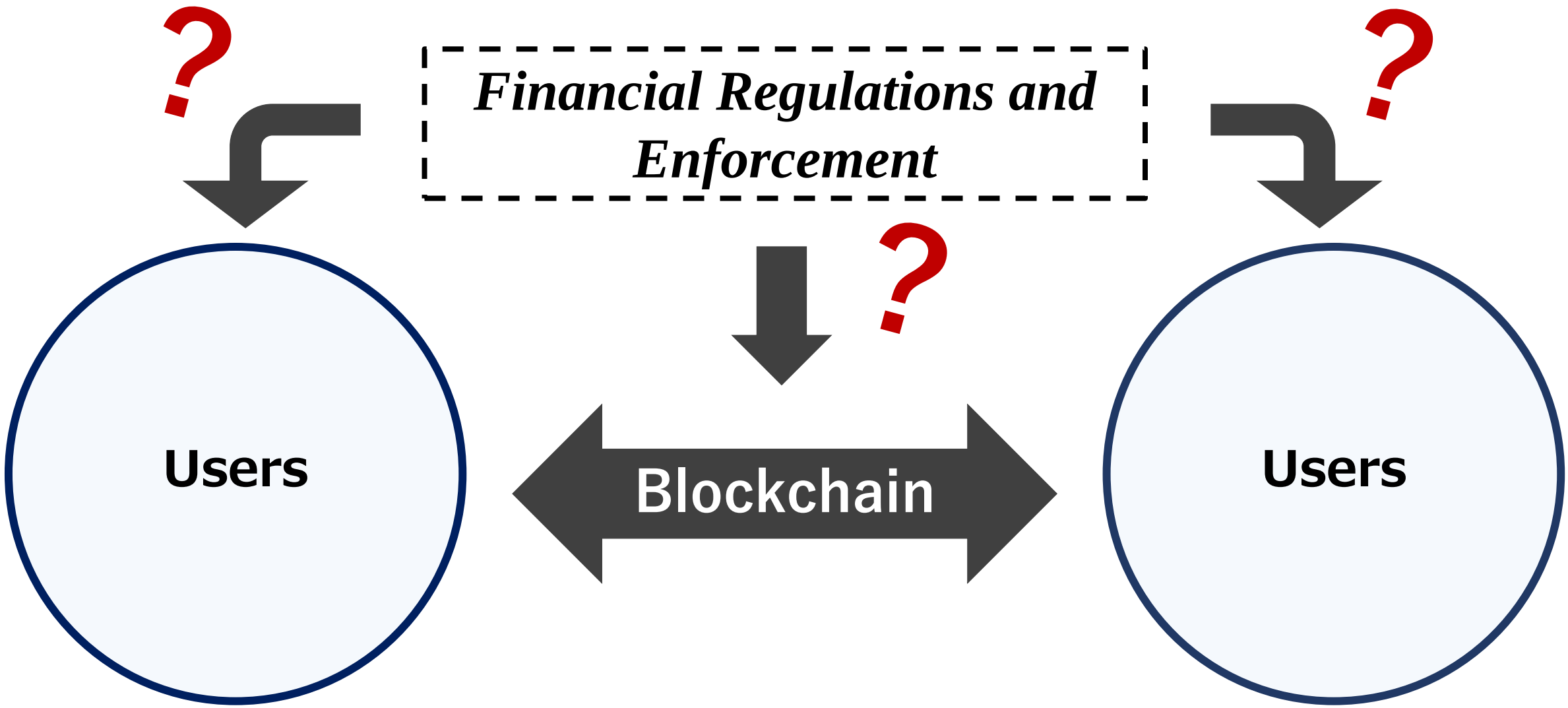
How can we achieve regulatory goals?



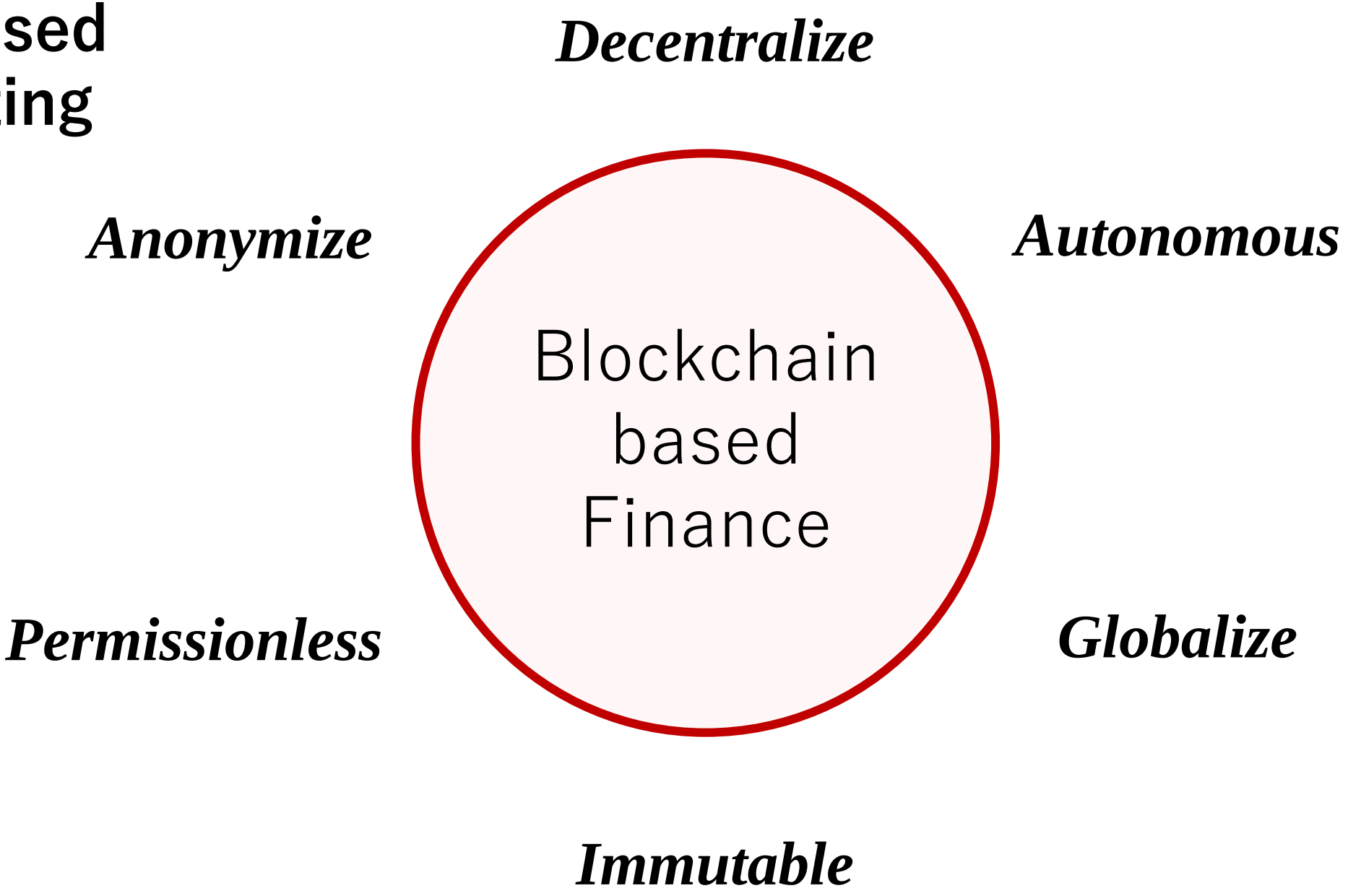
Current financial regulatory approach



Blockchain based finance and regulation



Six natures of blockchain based finance affecting regulation



Regulatability
＝Regulatory enforceability

Decentralized	*No regulatory target*
Autonomous	*Cannot stop the service*
Anonymized	*Reduced traceability*
Immutable	*No ex-post remedy*
Globalized	*Jurisdictional borders*
Permissionless	*Obscured responsibility*

Regulation alone may not be enough to achieve regulatory goals when the financial system becomes highly decentralized…

Emerging blockchain privacy enhancing technologies

Technology	Use Cases
Mixing	**Bitcoin**, Bitcoin Cash, Litecoin, Ethereum Dash etc.
Stealth Address	**Bitcoin** (Wallet), Monero
Ring Signature	Monero
Zero-Knowledge Proof (zk-SNARKs)	Zcash, Ethereum
Lightning Network	**Bitcoin**, Litecoin, Ethereum
Atomic Cross-Chain Swap	**Bitcoin**, Litecoin, Ethereum etc.
Mimblewimble	Grin, Beam
Schnorr Signature	Grin, Beam
Dandelion	Grin, Zcoin

Anonymity Benefits & Risks

Privacy

Fungibility

Cyber Crime

Cyber Terrorism

Security

Crypto-Laundering

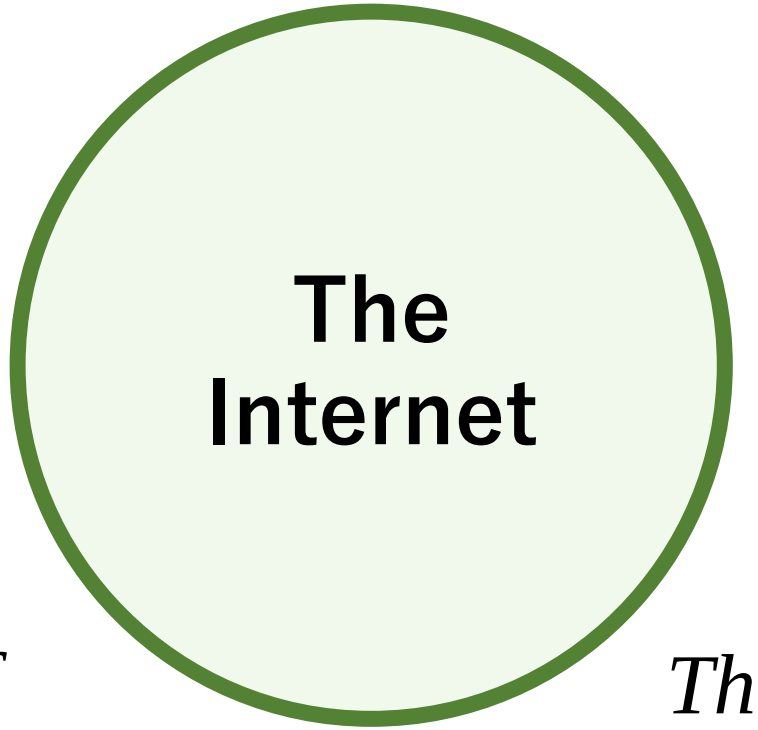
Censorship Resistance

Question we need to answer

- How could we continue to achieve financial regulatory goals as public interest?
- How should we resolve competing values?

Lessons from the Internet

Distributed control and new institutions



The large scale of communication

The global scope of the communication

History of governments and the Internet

Some governments tried to **put the Internet under the UN's control** in early 2000s.

Failed!!

Governments accepted the concept of ***Multi-stakeholder governance***

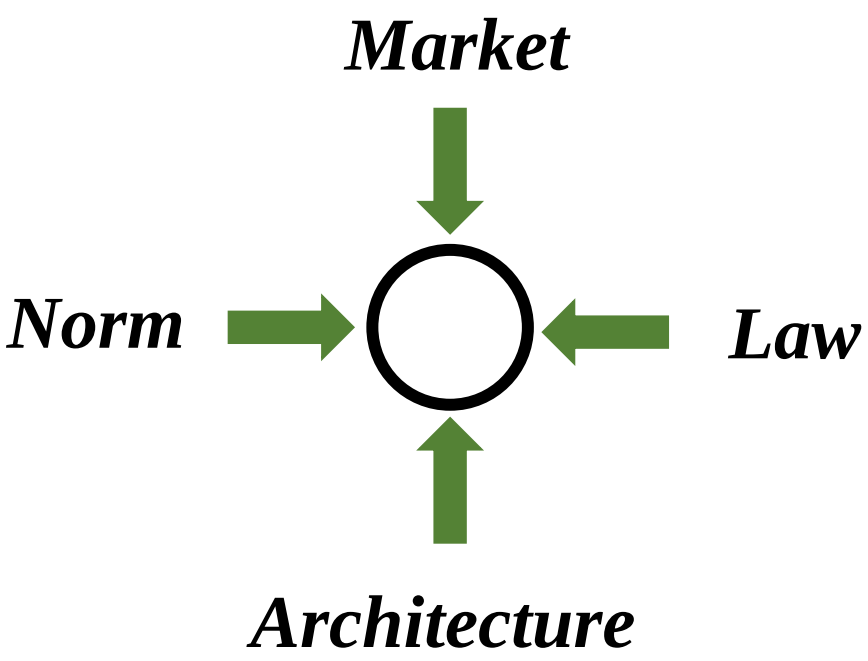
Social issues and multi-stakeholder cooperation

Intellectual property right protection; copyright and trademark

Contents regulation including child pornography

- Issues can't be solved by regulation alone
- Stakeholders have been cooperating to adderess them
- Need to overcome politics among stakeholders and within the stakeholder groups.

Classic governance model for cyberspace



Laurence Lessig – *Code*, 1997

Apply it to blockchain based finance

Law/regulation can be imposed on ecosystem participants such as minors

Government can intervene ***Market***

Government can join social activities to form desirable ***Norms***

Government can translate Law into Code (***Code as Law***)

De Filippi and Wright – *Blockchain and the Law* , 2018

CALL FOR MULTI-STAKEHOLDER COMMUNICATION TO ESTABLISH A GOVERNANCE MECHANISM FOR THE EMERGING BLOCKCHAIN-BASED FINANCIAL ECOSYSTEM

Yuta Takanashi, Shin'ichiro Matsuo, Eric Burger, Clare Sullivan, James Miller, Hirotoshi Sato*

ABSTRACT

Financial regulators around the world regulate financial intermediaries and activities to achieve their regulatory goals including investor/consumer protection, financial stability and prevention of financial crimes, and in so doing address various market failures. These objectives are needed in the social interest regardless of the technologies used by the financial system.

Blockchain technology and any financial ecosystem based on it have technical characteristics including decentralization, autonomization, anonymization and globalization, which could undermine the ability of regulators to achieve regulatory goals. Especially when it comes to preventing financial crimes, these characteristics could have significant negative impact on the ability of regulators. The intergovernmental Financial Action Task Force ("FATF")[1] recognizes these issues and is tackling them by issuing multiple guidelines; however, it seems that such

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Our idea

Develop ***Architectures/Codes*** that

- ✓ comply with ***Law/Regulation***
- ✓ align with ***Norm***
- ✓ is competitive in the ***Market***

Developing such architecture requires multi-stakeholders cooperation

Regulators should give impetus to develop multi-stakeholder governance

G20 in Fukuoka, Japan on June 8-9, 2019

Discussion by the G20 countries

FSB

Decentralised financial technologies *Report on financial stability, regulatory and governance implications (June, 2019)*

Decentralised financial technologies are likely to continue to evolve rapidly. Early liaison between regulators and a wider group of stakeholders might help ensure that **regulatory and other public policy objectives are considered in the initial design of technical protocols and applications**. This should help limit the emergence of unforeseen complications at a later stage.

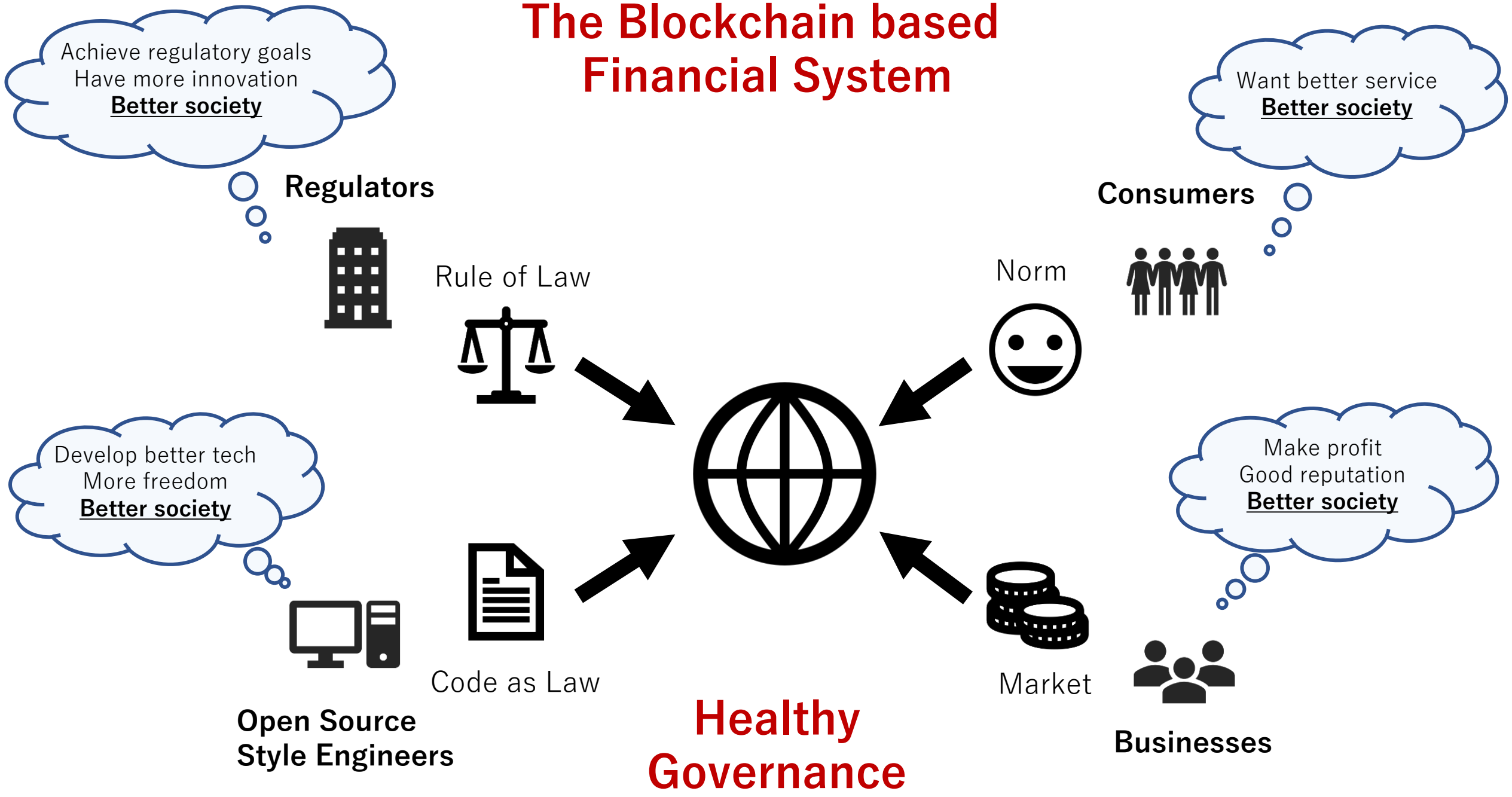
Authorities may therefore wish to enhance their dialogue and cooperation with a wider group of stakeholders, including software developers, the engineering community, as well as businesses, academia, and other relevant stakeholders such as investors, consumers and users. This would help to assess the opportunities and risks of decentralised financial technologies

G20 2019 JAPAN

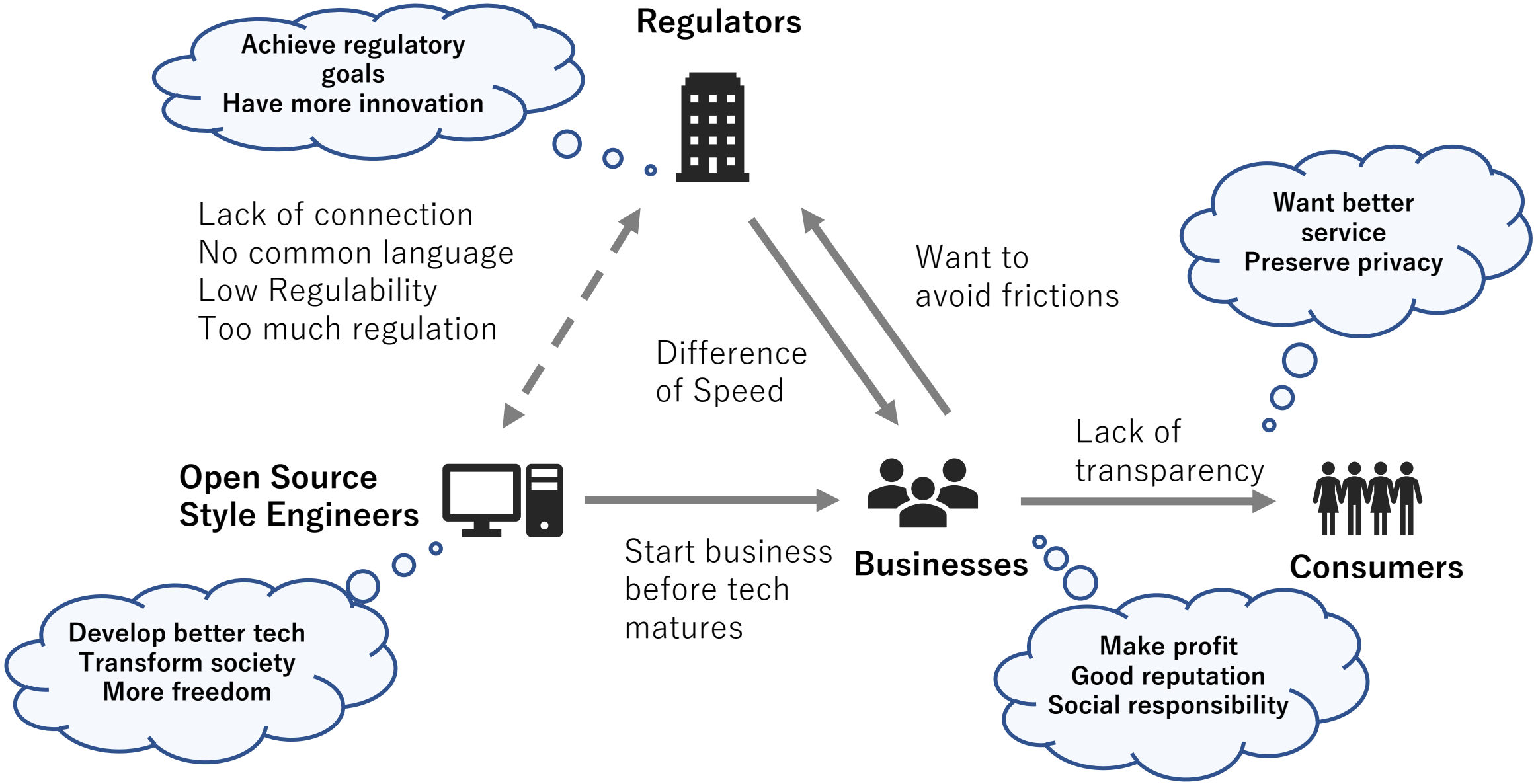
G20 Osaka Leaders' Declaration *(Jun, 2019)*

We also welcome the FSB report on decentralized financial technologies, and the possible implications for financial stability, regulation and governance, and **how regulators can enhance the dialogue with a wider group of stakeholders**.

The Blockchain based Financial System



Sounds easy?
Not really…



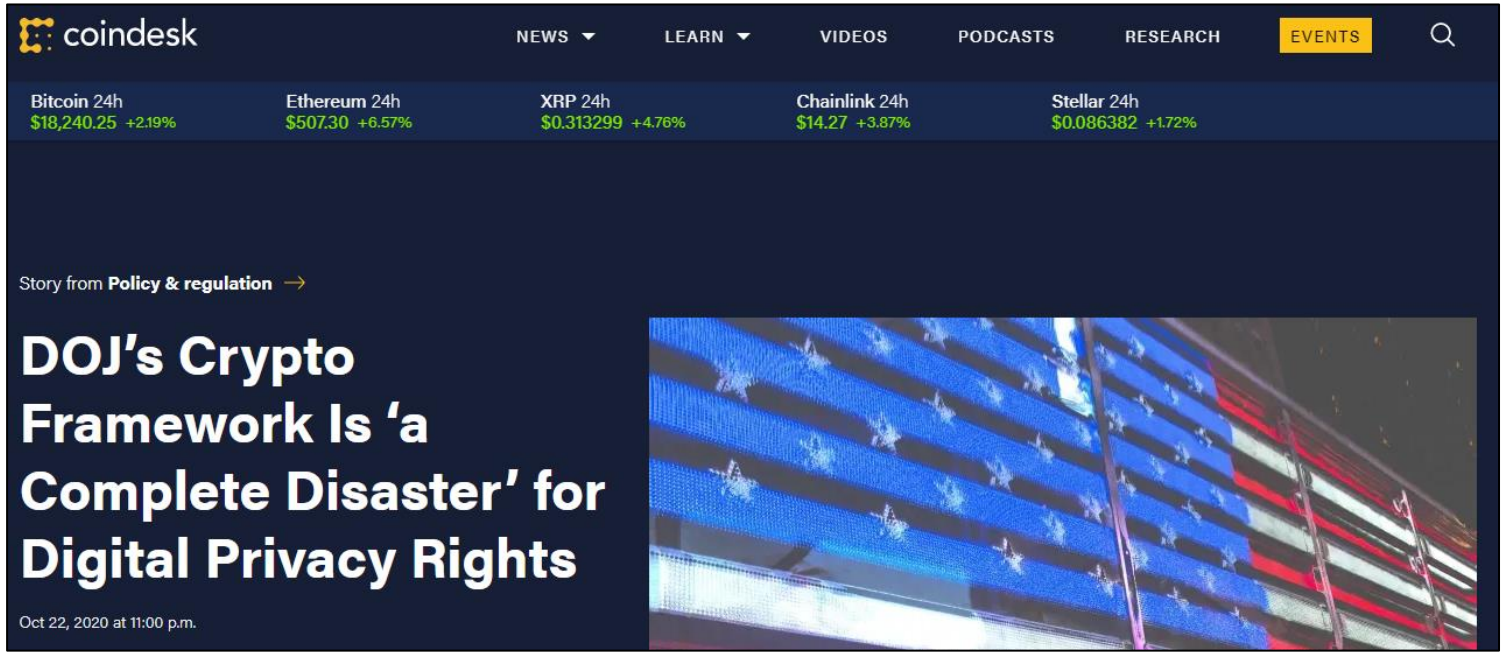
Surging regulatory pressure…

G7 FMCBG Statement on Digital Payments
October, 2020

the G7 is concerned by the rising threat of ransomware attacks, ….
These attacks, which often involve payments in crypto-assets, jeopardize essential functions along with our collective security and prosperity. We affirm our **resolve to combat this threat** collectively as well as individually.

FATF Report to the G20 FMCBG on So-called Stablecoins *June,2020*

The FATF has also identified potential risks which **may require further action**, …so-called stablecoins with **decentralised governance structures** … and **anonymous peer-to-peer transactions via unhosted wallets**



What issues should we solve? – We are not enemies but friends!

No coordination

- All stakeholders aim better society.
- But fail to coordinate

Misunderstandings

- Absence of common language and place for communication
- Causes lots of misunderstandings

Blockers for innovation

Independent works may produce

- Incompatibility to social order
- Slow down permissionless innovation

Let's work together at BGIN for the social good and the bright future!